

Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Kalman Filter

Nov 12, 2025

Welcome to the lecture!

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Technische Hochschule Ingolstadt

AIMotion, AI and Sensor Data Fusion

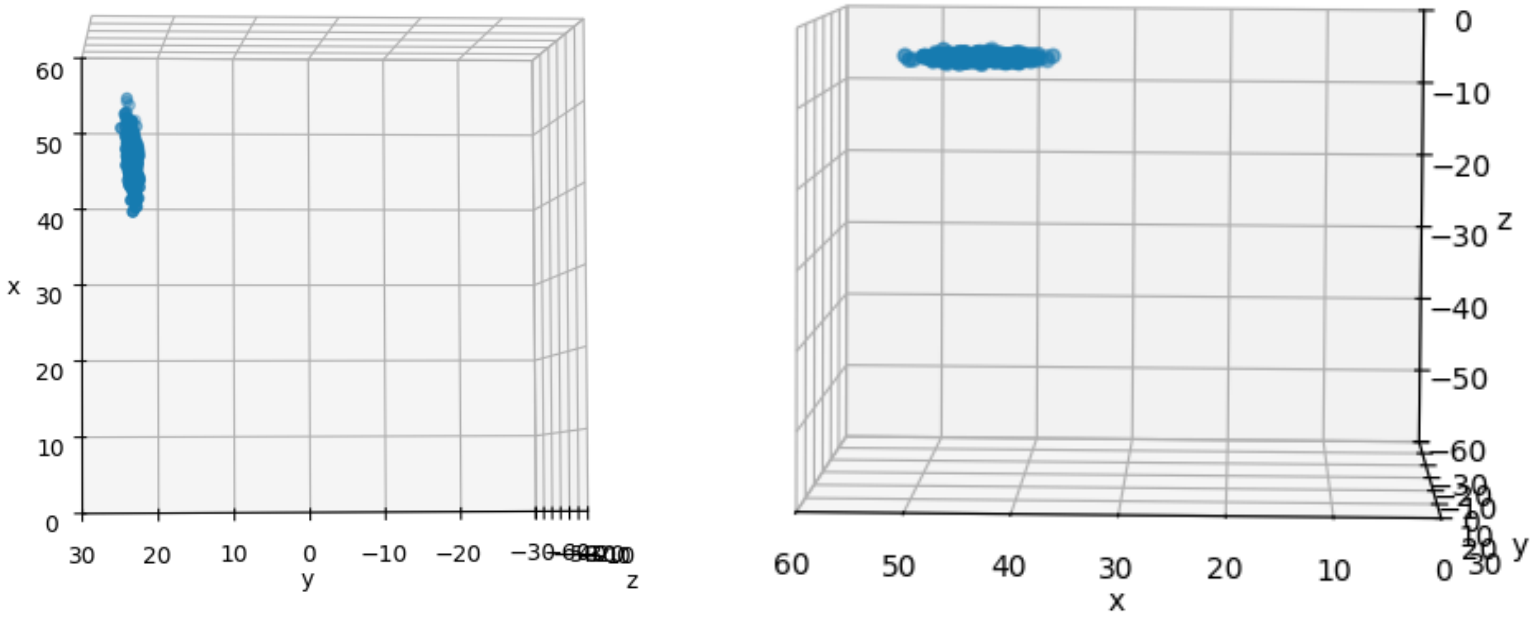
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- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

Today's Lecture Goals --

- **Discuss Sheet 2 and Programming Exercises**
- **Calculating a Bayesian Inference Example**
- **And then we will finally go into the Kalman Filter.**
  - First the linear variant – Deriving the equations and motivating the new exercise sheet.
  - We will also check how the Kalman Filter is an optimal Bayesian estimator.
  - In the next lectures, we will look at nonlinear extensions –
    the Extended and the Unscented Kalman Filters.
  - We will finally wrap up with using Kalman Filters for Multi-Object Tracking (MOT) applications.

Exercise Sheet 2
Programming Exercises



Covariance scatter plots for the 3D exercise

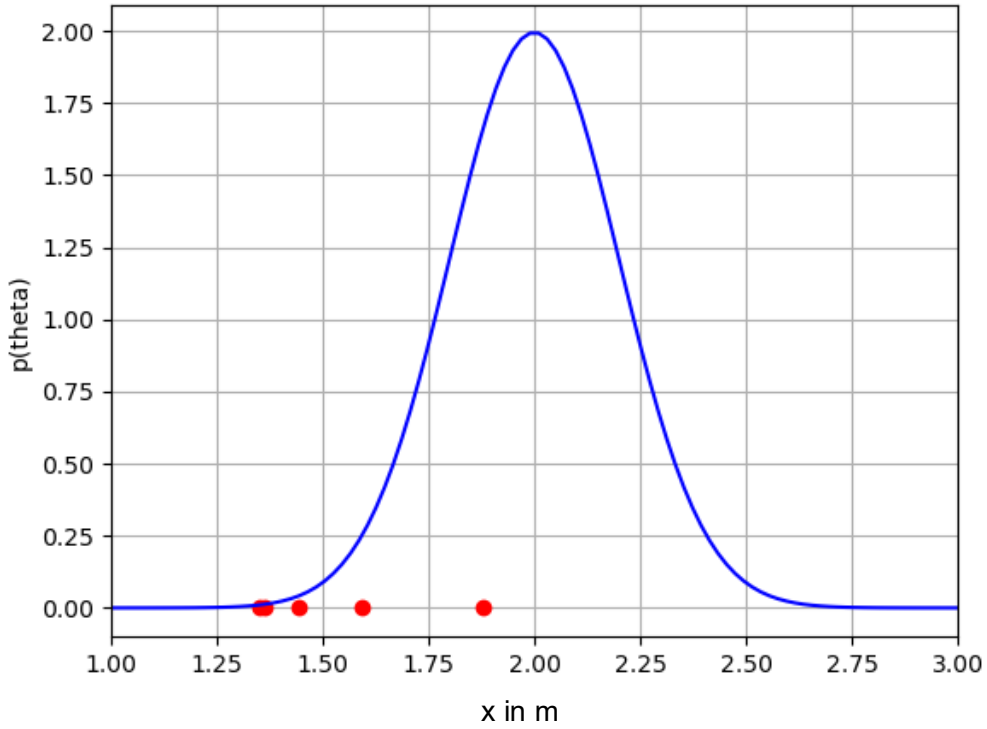
Bayesian Inference Example

Application Example: Bayesian Inference

- We measure the width of an object and collected 5 measurements:

$$y_1 = 1.88m, \quad y_2 = 1.36m, \quad y_3 = 1.59m$$
$$y_4 = 1.35m, \quad y_5 = 1.44m$$

- We assume, the measurement errors are normally distributed,
  But we do not know the parameters of the distribution
- Prior information: The width of the object is normally distributed
  $\mu_p = 2\,m$ und $\sigma_p^2 = 0{,}04\,m^2$
- Objective: Derive an estimate x of the width combining (fusing) the measured with the prior information



Prior (blue) and measurements (red)

Application Example: Bayesian Inference - 1

- We apply Bayes rule and compute

$$p(x|\boldsymbol{y}) \propto p(\boldsymbol{y}|x) \cdot p(x)$$

- With Gaussian probability density functions for **prior** and **likelihood**

$$p(x) = \frac{1}{\sigma_p\sqrt{2\pi}} e^{\frac{(x-\mu_p)^2}{2\sigma_p^2}} \quad \text{and} \quad p(y_i|\boldsymbol{x}) = \mathcal{N}(\mu_L, \sigma_L^2) = \frac{1}{\sigma_L\sqrt{2\pi}} e^{-\frac{(y_i-\mu_L)^2}{2\sigma_L^2}}$$

- For the product, we get

$$p(x|\boldsymbol{y}) = \frac{1}{\sqrt{2\pi}\sigma_{post}} \cdot \exp\left(-\frac{(x-\mu_{post})^2}{2\sigma_{post}^2}\right) \cdot \frac{1}{\sqrt{2\pi}\cdot(\sigma_p^2+\sigma_L^2)} \cdot \exp\left(-\frac{(\mu_p+\mu_L)^2}{2(\sigma_p^2+\sigma_L^2)}\right)$$

with

$$\sigma_{post} = \sqrt{\frac{\sigma_L^2\sigma_p^2}{\sigma_p^2+\sigma_L^2}} \qquad \mu_{post} = \frac{\mu_p\sigma_L^2 + \mu_L\sigma_p^2}{\sigma_p^2+\sigma_L^2}$$



We need the parameters μ_L and σ_L of the Likelihood-function $p(y|x)$

Application Example: Bayesian Inference - 2

- Determine the parameters μ_L and σ_L of the Likelihood-function by means of the **Maximum Likelihood method,** we maximize the probability density, taking into account the obtained measurements

$$p(\boldsymbol{y}|x) = \frac{1}{\sigma_L\sqrt{2\pi}} e^{\frac{(1.88-\mu_L)^2}{2\sigma_L^2}} \cdot \frac{1}{\sigma_L\sqrt{2\pi}} e^{\frac{(1.36-\mu_L)^2}{2\sigma_L^2}} \cdot \frac{1}{\sigma_L\sqrt{2\pi}} e^{\frac{(1.59-\mu_L)^2}{2\sigma_L^2}} \cdot \frac{1}{\sigma_L\sqrt{2\pi}} e^{\frac{(1.35-\mu_L)^2}{2\sigma_L^2}} \cdot \frac{1}{\sigma_L\sqrt{2\pi}} e^{\frac{(1.44-\mu_L)^2}{2\sigma_L^2}}$$

- Trick: Employing the log-Likelihood simplifies the computations

$$\begin{aligned}\ln\left(p(\boldsymbol{y}|x)\right) &= 5 \cdot \ln\left(\frac{1}{\sigma_L\sqrt{2\pi}}\right) - \frac{(1.88-\mu_L)^2}{2\sigma_L^2} - \frac{(1.36-\mu_L)^2}{2\sigma_L^2} - \frac{(1.59-\mu_L)^2}{2\sigma_L^2} \\ &\quad - \frac{(1.35-\mu_L)^2}{2\sigma_L^2} - \frac{(1.44-\mu_L)^2}{2\sigma_L^2} \\ &= -5\ln(\sigma_L) + \frac{5}{2}\ln(2\pi) - \frac{(1.88-\mu_L)^2}{2\sigma_L^2} - \frac{(1.36-\mu_L)^2}{2\sigma_L^2} - \frac{(1.59-\mu_L)^2}{2\sigma_L^2} \\ &\quad - \frac{(1.35-\mu_L)^2}{2\sigma_L^2} - \frac{(1.44-\mu_L)^2}{2\sigma_L^2}\end{aligned}$$

Application Example: Bayesian Inference - 3

- Maximizing the log-Likelihood

$$\begin{aligned}\ln\left(p(\boldsymbol{y}|x)\right) &= 5 \cdot \ln\left(\frac{1}{\sigma_L\sqrt{2\pi}}\right) - \frac{(1.88-\mu_L)^2}{2\sigma_L^2} - \frac{(1.36-\mu_L)^2}{2\sigma_L^2} - \frac{(1.59-\mu_L)^2}{2\sigma_L^2} \\ &\quad - \frac{(1.35-\mu_L)^2}{2\sigma_L^2} - \frac{(1.44-\mu_L)^2}{2\sigma_L^2} \\ &= -5\ln(\sigma_L) + \frac{5}{2}\ln(2\pi) - \frac{(1.88-\mu_L)^2}{2\sigma_L^2} - \frac{(1.36-\mu_L)^2}{2\sigma_L^2} - \frac{(1.59-\mu_L)^2}{2\sigma_L^2} \\ &\quad - \frac{(1.35-\mu_L)^2}{2\sigma_L^2} - \frac{(1.44-\mu_L)^2}{2\sigma_L^2}\end{aligned}$$

- The parameters result from the conditions at maximum points

$$\frac{\partial \ln\left(p(\boldsymbol{y}|x)\right)}{\partial \mu_L} = 0$$
$$\frac{\partial \ln\left(p(\boldsymbol{y}|x)\right)}{\partial \sigma_L} = 0$$

Application Example: Bayesian Inference - 4

- We obtain

$$\frac{\partial \ln\left(p(\boldsymbol{y}|x)\right)}{\partial \mu_L} = \frac{1}{2\sigma_L^2}\left[(1.88-\mu_L)+(1.36-\mu_L)+(1.59-\mu_L)+(1.35-\mu_L)+(1.44-\mu_L)\right] = 0$$

$$\Leftrightarrow \mu_L = \frac{1.88+1.36+1.59+1.35+1.44}{5} = 1.53$$

and

$$\begin{aligned}\frac{\partial \ln\left(p(\boldsymbol{y}|x)\right)}{\partial \sigma_L} &= -\frac{5}{\sigma_L} + \sigma_L^{-3}\cdot\big((1.88-\mu_L)^2+(1.36-\mu_L)^2 \\ &\quad +(1.59-\mu_L)^2+(1.35-\mu_L)^2+(1.44-\mu_L)^2\big) = 0 \\ &= -\frac{5}{\sigma_L} + 0.19\cdot\sigma_L^{-3}\end{aligned}$$

$$\Leftrightarrow \sigma_L^2 = \frac{0.19}{5}$$

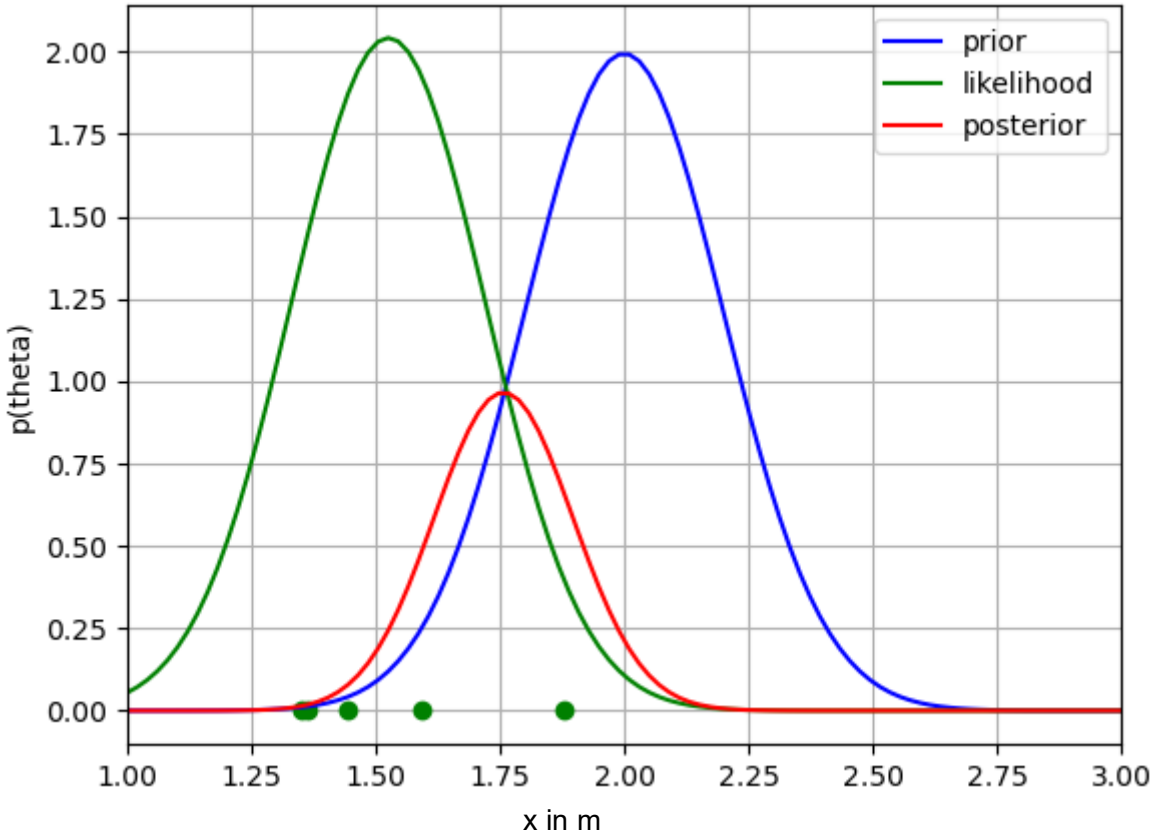
Application Example: Bayesian Inference - 5

- Now we are able to compute

$$p(x|\boldsymbol{y}) \propto p(\boldsymbol{y}|x) \cdot p(x)$$

- We obtain the maximum of the posterior probability density function from ($\mu_{post} = 1.75$)

$$\mu_{post} = \frac{\mu_p \sigma_L^2 + \mu_L \sigma_p^2}{\sigma_p^2 + \sigma_L^2}$$



Please note: $p(\theta|D)$ is not normalized, $p(D)$ was not computed!

Kalman Filter

Let‘s assume we want to determine the motion of an object moving in a straight line.

Let‘s also assume we want to model its position, velocity, and acceleration.

- **The object‘s system state at time *t* can be expressed as** $\mathbf{x}_t = \begin{bmatrix} s_t \\ v_t \\ a_t \end{bmatrix}$.

- **As time progresses, the object‘s state changes according to a model:**

$$\left.\begin{aligned} s_{t+1} &= s_t + \Delta t \cdot v_t + \frac{1}{2}(\Delta t)^2 \cdot a_t \\ v_{t+1} &= v_t + \Delta t \cdot a_t \\ a_{t+1} &= a_t \end{aligned}\right\} \quad \mathbf{x}_{t+1} = \mathbf{A} \cdot \mathbf{x}_t \quad \textbf{with} \quad A = \begin{bmatrix} 1 & \Delta t & \frac{1}{2}(\Delta t)^2 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{bmatrix}.$$

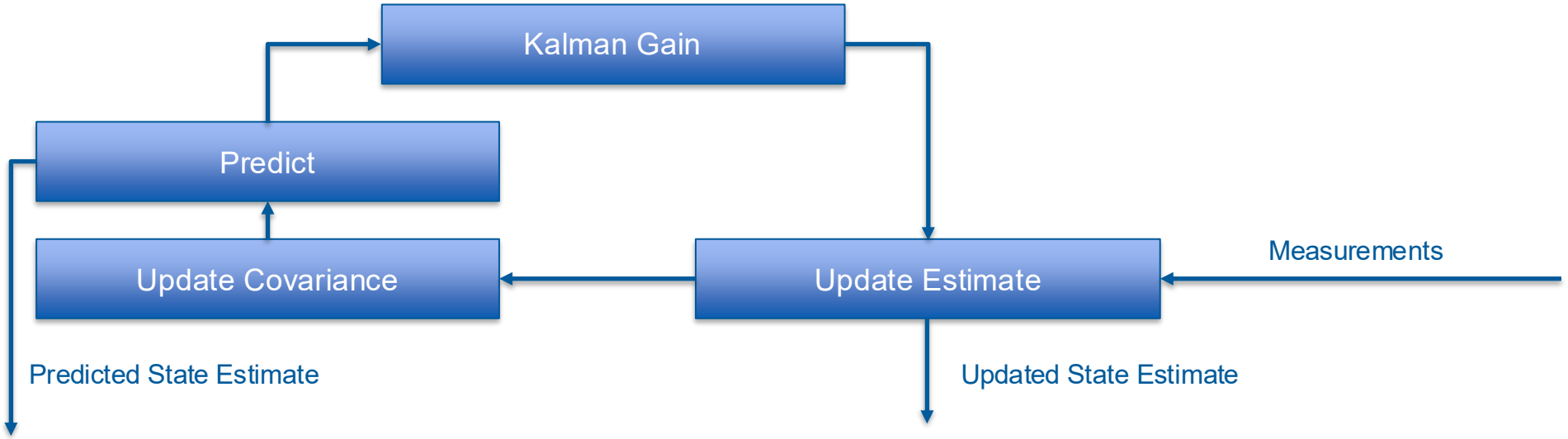


Let‘s also assume we measure the object with a sensor, let‘s say a radar, that can measure distance and velocity but not acceleration:

$$\hat{\mathbf{z}}_t = \begin{bmatrix} \hat{s}_t \\ \hat{v}_t \end{bmatrix}$$

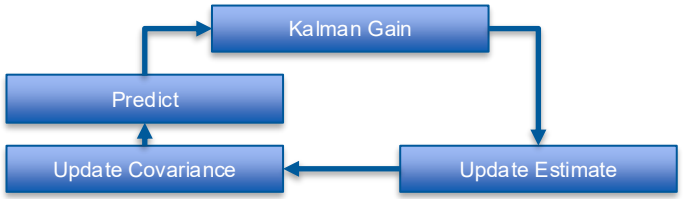
We can express an estimate of what *z* should look like by creating a measurement matrix H:

$$\mathbf{z}_t = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} s_t \\ v_t \end{bmatrix} = \mathbf{H}\mathbf{x}_t$$



Equations:

- Kalman Gain $K_t = P_t' H^\top (H P_t' H^\top + R)^{-1}$
- Update Estimate $\hat{\mathbf{x}}_t = \mathbf{x}_t' + K_t(\mathbf{z}_t - H\hat{\mathbf{x}}_t')$
- Update Covariance $P_t = (I - K_t H) P_t'$
- Predict $\mathbf{x}_{t+1}' = A\hat{\mathbf{x}}_t$
  $P_{t+1} = A P_t A^\top + Q$



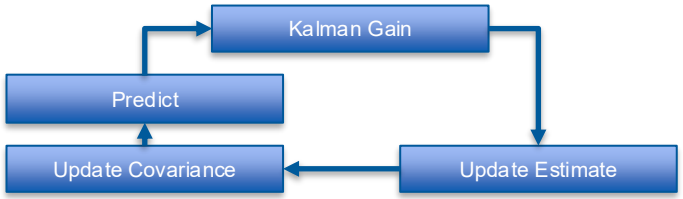
- **We expect to make an error both in**
  - updating the model (because of inaccuracies in the model itself, and previous errors), and
  - measuring (because of noise and other residual errors).
- **Let‘s introduce two random variables $\mathbf{v}_t$ (nx1, same dimensions as the state) and $\mathbf{w}_t$ (mx1, same dimensions as the measurement). We assume they are normal distributed and will use that later.**

$$\hat{\mathbf{x}}_{t+1} = \mathbf{A} \cdot \mathbf{x}_t + \mathbf{w}_t$$
$$\hat{\mathbf{z}}_{t+1} = \mathbf{H} \cdot \hat{\mathbf{x}}_{t+1} + \mathbf{v}_t$$

- **Our task is to optimize our system state estimation $\mathbf{x}_t$, which we can do with an error or loss function $\mathcal{L}$. Let‘s choose the mean squared error function**

$$\mathcal{L}(\hat{\mathbf{x}}, \mathbf{x}) = E[\sum (\hat{x}_i - x_i)^2]$$

(why can we do that?)

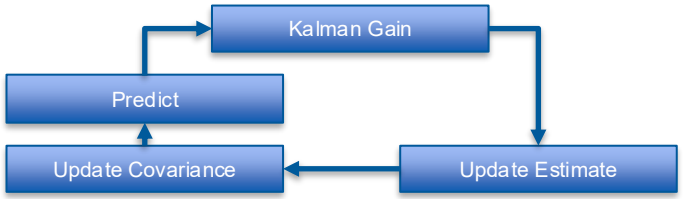


Maximum Likelihood Estimation

- We are looking for a ground truth value of x,
- for which we have an estimate of $\hat{x}$,
- but can only measure $y = x + w$
  (1-dimensional for simplicity).
- We assume that w is normal distributed with an expectation of 0 and variance of σ^2, so y is normal distributed with an expectation of $\hat{x}$ and variance of σ^2:

$$P(y|\hat{x}) \propto e^{-\frac{(y-\hat{x})^2}{2\sigma^2}}$$

(we're leaving out the constant norming term at the beginning).



Maximum Likelihood Estimation

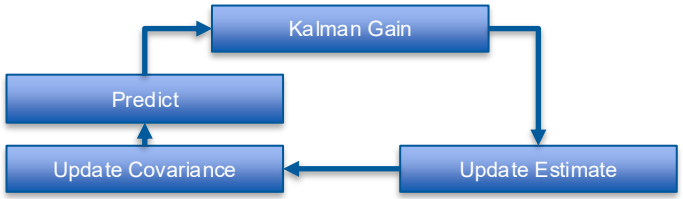
- The likelihood of y being observed over all measurements is

$$P(\mathbf{y}|\mathbf{x}) = P(y_1|\hat{x}_i \cap y_2|\hat{x}_2 \cap \cdots \cap y_t|\hat{x}_t)$$
$$= \prod_t P(y_t|\hat{x}_t)$$
$$= \prod_t e^{-\frac{(y_t-\hat{x}_t)^2}{2\sigma^2}}$$

- We want to find the $\hat{x}$ that maximizes the likelihood of the observation .
  We can make our lives a lot simpler though – any max of $f(x)$ is also a max of $\log f(x)$, so instead we can look for the max of

$$\log P(\mathbf{y}|\hat{\mathbf{x}}) = -\frac{1}{2}\sum_t \left(\frac{(y_t - \hat{x}_t)^2}{\sigma_t^2}\right)$$

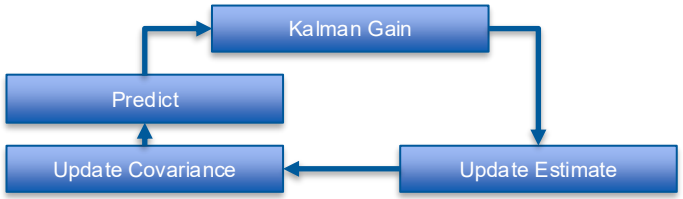
- The core of this log-likelihood function is the mean squared error.



- **We expect to make an error both in**
  - updating the model (because of inaccuracies in the model itself, and previous errors), and
  - measuring (because of noise and other residual errors).
- **Let‘s introduce two random variables $\mathbf{w}_t$ (dimensions $n \times 1$, same as the state) and $\mathbf{v}_t$ ($m \times 1$, same dimensions as the measurement).**

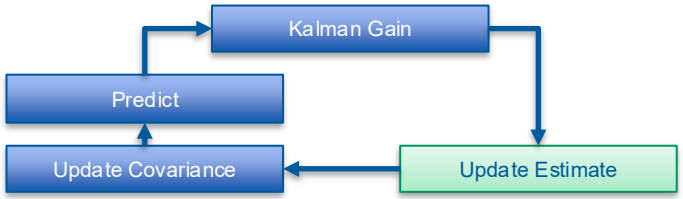
$$\hat{\mathbf{x}}_{t+1} = \mathbf{A} \cdot \mathbf{x}_t + \mathbf{w}_t$$
$$\hat{\mathbf{z}}_{t+1} = \mathbf{H} \cdot \hat{\mathbf{x}}_{t+1} + \mathbf{v}_t$$

- **We make the following assumptions:**
  - $\mathbf{A}$ is noiseless and may be different from observation to observation (e.g. because of small time jitter)
  - $\mathbf{H}$ is noiseless and can be assumed constant for the simple case
  - $\mathbf{w}_t$ and $\mathbf{v}_t$ are normal distributed with zero mean.



Noise Models

- **To characterize the noise models, we introduce the covariance matrices**
  - $Q = E[\mathbf{w}\mathbf{w}^\top]$ for the model noise, i.e. $\mathbf{w} \sim N(\mathbf{0}, Q)$ (shorthand for „w is is normal distributed with zero mean and Q variance“), and
  - $R = E[\mathbf{v}\mathbf{v}^\top]$ for the measurement noise, i.e. $\mathbf{v} \sim N(\mathbf{0}, R)$.
- **We can characterize the distribution of the mean squared error as** $P_t = E\left[(\mathbf{x}_t - \hat{\mathbf{x}}_t)(\mathbf{x}_t - \hat{\mathbf{x}}_t)^\top\right]$.



Estimate Update

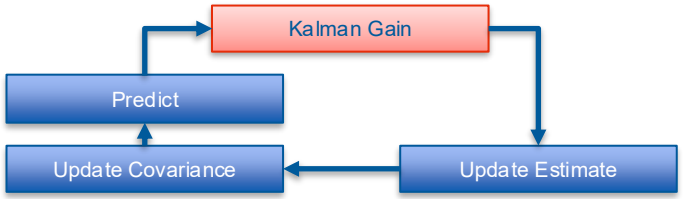
- Let's assume the prior estimate for $\hat{\mathbf{x}}_t$ is called $\hat{\mathbf{x}}'_t$: $\hat{\mathbf{x}}'_t = A\hat{\mathbf{x}}_{t-1}$.
- From that prior estimate, we know our measurement *should be* $\hat{\mathbf{z}}'_t = H\hat{\mathbf{x}}'_t$
- We obtain the *innovation* from comparing that prior estimate with our *actual* measurement $\mathbf{z}_t$:

$$\mathbf{i}_t = \mathbf{z}_t - \hat{\mathbf{z}}'_t = \mathbf{z}_t - H\hat{\mathbf{x}}'_t$$

- With the innovation, we can write the equation to update the estimate as

$$\begin{aligned}\hat{\mathbf{x}}_t &= \hat{\mathbf{x}}'_t + K_t(\mathbf{i}_t) \\ &= \hat{\mathbf{x}}'_t + K_t(\mathbf{z}_t - H\hat{\mathbf{x}}'_t)\end{aligned}$$

- K is called the *Kalman Gain*, and we will derive it now.



Deriving the Kalman Gain – Step 1

- With $\mathbf{z}_t = H\mathbf{x}_t + \mathbf{v}_t$, we have

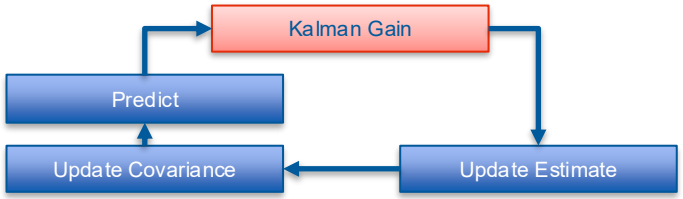
$$\hat{\mathbf{x}}_t = \hat{\mathbf{x}}'_t + K_t(H\mathbf{x}_t + \mathbf{v}_t - H\hat{\mathbf{x}}'_t)$$

which we can write as

$$\begin{aligned}\mathbf{x}_t - \hat{\mathbf{x}}_t &= \mathbf{x}_t - \hat{\mathbf{x}}'_t - K_t(H\mathbf{x}_t + \mathbf{v}_t - H\hat{\mathbf{x}}'_t) \\ &= \mathbf{x}_t - K_tH\mathbf{x}_t - \hat{\mathbf{x}}'_t + K_tH\hat{\mathbf{x}}'_t - K\mathbf{v}_t \\ &= (I - K_tH)(\mathbf{x}_t - \hat{\mathbf{x}}'_t) - K_t\mathbf{v}_t\end{aligned}$$

and with $P_t = E\left[(\mathbf{x}_t - \hat{\mathbf{x}}_t)(\mathbf{x}_t - \hat{\mathbf{x}}_t)^\top\right]$, we can expand to

$$\begin{aligned}P_t = E[&[(I - K_tH)(\mathbf{x}_t - \hat{\mathbf{x}}'_t) - K_t\mathbf{v}_t] \\ &[(I - K_tH)(\mathbf{x}_t - \hat{\mathbf{x}}'_t) - K_t\mathbf{v}_t]^\top]\end{aligned}$$



Deriving the Kalman Gain – Step 2

- In the equation $$P_t = E[[(I - K_t H)(\mathbf{x}_t - \hat{\mathbf{x}}'_t) - K_t \mathbf{v}_t] \\ [(I - K_t H)(\mathbf{x}_t - \hat{\mathbf{x}}'_t) - K_t \mathbf{v}_t]^\top]$$

  $\mathbf{x}_t - \mathbf{x}'_t$ is the error of the prior estimate. This is uncorrelated to the measurement noise, so we can write the expectation as

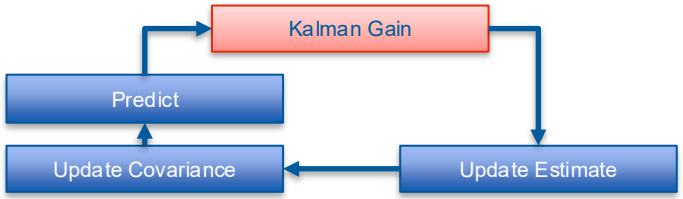
  $$P_t = (I - K_t H) E\left[(\mathbf{x}_t - \hat{\mathbf{x}}'_t)(\mathbf{x}_t - \hat{\mathbf{x}}'_t)^\top\right] (I - K_t H)^\top \\ + K_t E[\mathbf{v}_t \mathbf{v}_t^\top] K_t \top$$

  and with the above introduction of $R = E[\mathbf{v}_t \mathbf{v}_t^\top]$ and $P_t = E\left[(\mathbf{x}_t - \hat{\mathbf{x}}'_t)(\mathbf{x}_t - \hat{\mathbf{x}}'_t)^\top\right]$, as

  $$P_t = (I - K_t H) P'_t (I - K_t H)^\top + K_t R K_t^\top$$

  where P'_t is the prior estimate of P_t .

- This equation is the error update equation.



Deriving the Kalman Gain – Step 3

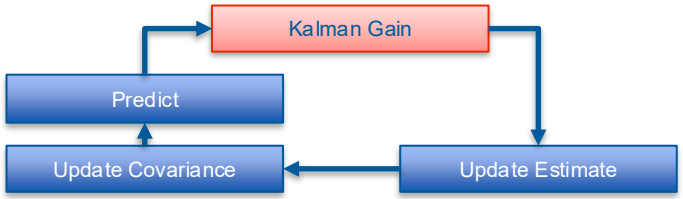
- **Error Update Equation:**

$$P_t = (I - K_t H) P'_t (I - K_t H)^\top + K_t R K_t^\top$$

Transform — Previous estimate — Measurement noise

Interpolation (1-gain)*x + gain*y

- So far, we have „handwaved“ around K – now we can derive the optimal K to optimize the error update equation.
- This means minimizing the error update equation for the elements of K.
- The diagonal elements of $P_t = E\left[(\mathbf{x}_t - \hat{\mathbf{x}}'_t)(\mathbf{x}_t - \hat{\mathbf{x}}'_t)^\top\right]$ are the mean squared errors, and their sum is the *trace* of the matrix.
- So we can minimize by minimizing the trace of P_t, i.e. differentiating by K_t and setting to zero.



Deriving the Kalman Gain – Step 4

- **Expanding** $P_t = (I - K_t H) P'_t (I - K_t H)^\top + K_t R K_t^\top$ **gives**

$$\begin{aligned} P_t =& (P'_t - K_t H P'_t)(I - K_t H)^\top + K R K^\top \\ =& P'_t - K_t H P'_t - P'(K_t H)^\top + K_t H P' t (K_t H)^\top + K R K^\top \\ =& P'_t - K_t H P'_t - P'_t (K_t H)^\top + K_t (H P'_t H^\top + R) K_t^\top \\ =& P'_t - 2 P'_t (K_t H)^\top + K_t (H P'_t H^\top + R) K_t^\top . \end{aligned}$$

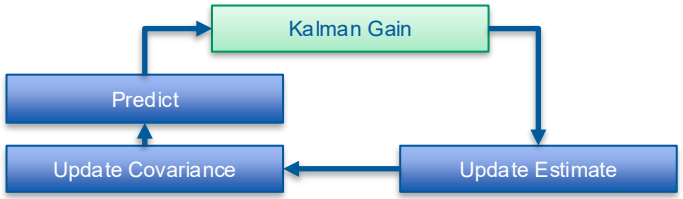
- **Pulling the trace over this gives:**

$$T[P_t] = T[P'_t] - 2T[P'_t (K_t H)^\top] + T[K_t (H P'_t H^\top + R) K_t^\top].$$

- **Differentiating with respect to K_t gives**

$$\frac{dT[P_t]}{dK_t} = -2T[P'_t H^\top] + 2T[K_t (H P'_t H^\top + R)] = 0.$$

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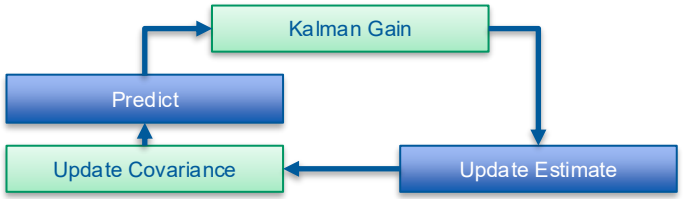


Deriving the Kalman Gain – Step 5

- **Setting** $\frac{dT[P_t]}{dK_t} = 0$ **, we get**

$$-2P_t'H^\top + 2K_t(HP_t'H^\top + R) = 0 \Rightarrow$$
$$P_t'H^\top = K_t(HP_t'H^\top + R) \Rightarrow$$
$$K_t = P_t'H^\top(HP_t'H^\top + R)^{-1}.$$

- $K_t = P_t'H^\top(HP_t'H^\top + R)^{-1}$ **is the Kalman Gain Equation.**

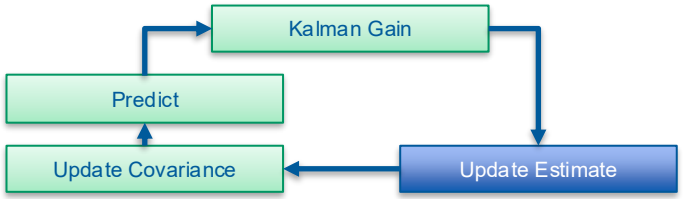


- **With the Kalman gain, we can update the prediction covariance.**
- **Taking** $P_t = P'_t - K_t H P'_t - P'_t (K_t H)^\top + K_t (H P'_t H^\top + R) K_t^\top$ **from above and substituting**

$$K_t = P'_t H^\top (H P'_t H^\top + R)^{-1}$$ **, we get (derivation on blackboard)**

$$\begin{aligned} P_t &= P'_t - P'_t H^\top (H P'_t H^\top + R)^{-1} H P'_t \\ &= P'_t - K_t H P'_t \\ &= (I - K_t H) P'_t \end{aligned}$$

- **This gives us the formula for the error covariance update.**

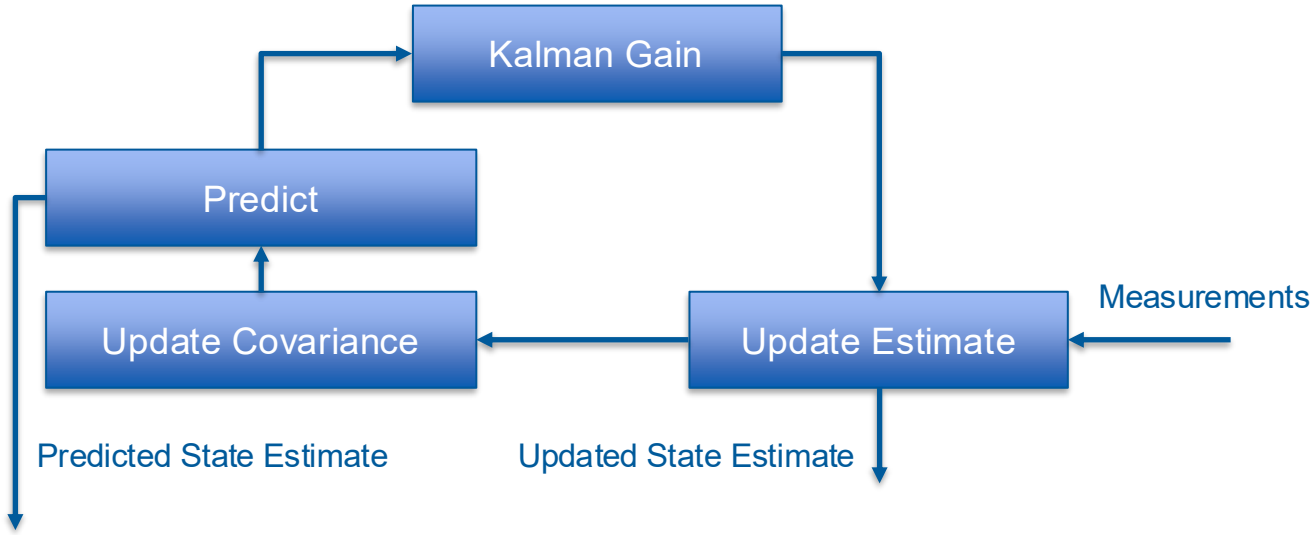


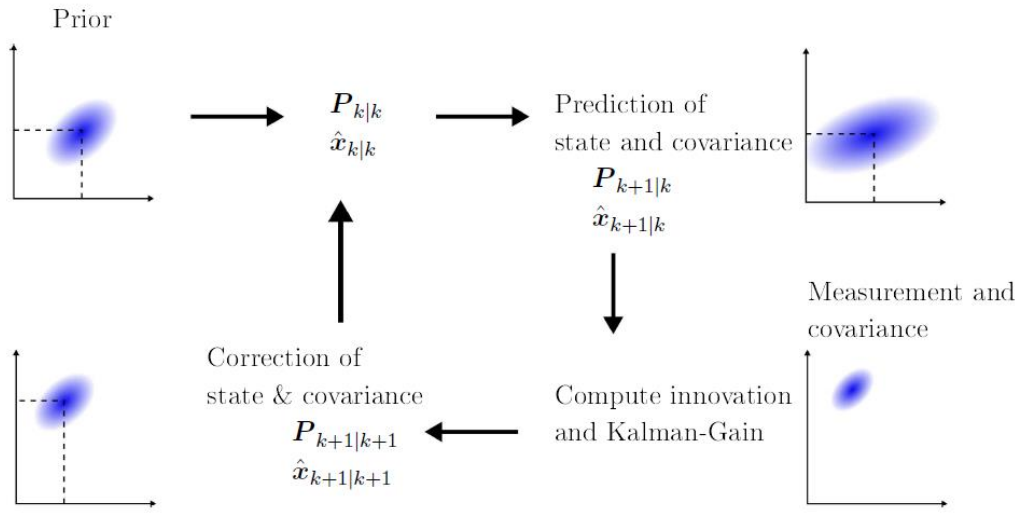
Predicted State, Prediction Covariance

- **The state prediction is given by** $\mathbf{x}'_{t+1} = A\hat{\mathbf{x}}_t$.
- **To predict the error covariance into t+1, we form an expression for the prior error:**

$$\begin{aligned}\mathbf{e}'_{t+1} &= \mathbf{x}_{t+1} - \hat{\mathbf{x}}'_{t+1} \\ &= (A\mathbf{x}_t + \mathbf{w}_t) - A\hat{\mathbf{x}}_t \\ &= A(\mathbf{x}_t - \hat{\mathbf{x}}_k) + w_t \\ &= A\mathbf{e}_t + w_t\end{aligned}$$

- **With** $P'_t = E[\mathbf{e}'_t\mathbf{e}'^{\top}_t]$ **, we have** $\begin{aligned}P'_{t+1} &= E[\mathbf{e}'_{t+1}\mathbf{e}'^{\top}_{t+1}] \\ &= E\left[A\mathbf{e}_t(A\mathbf{e}_t)^{\top}\right] + E[\mathbf{w}_t\mathbf{w}_t^{\top}] \\ &= AP_tA^{\top} + Q\end{aligned}$
- **This completes the recursion.**





Iterations of the Kalman Filter

Equations:

- Kalman Gain $K_t = P'_t H^\top (H P'_t H^\top + R)^{-1}$
- Update Estimate $\hat{\mathbf{x}}_t = \mathbf{x}'_t + K_t(\mathbf{z}_t - H\hat{\mathbf{x}}'_t)$
- Update Covariance $P_t = (I - K_t H)P'_t$
- Predict $\mathbf{x}'_{t+1} = A\hat{\mathbf{x}}_t$

  $P_{t+1} = AP_tA^\top + Q$



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Thank you very much!